

TFT-LCD Panel Fundamentals

Thin-film transistor liquid-crystal displays (TFT-LCDs) are a variant of LCDs that offer improved image quality. Find out what goes into their making

■ DANIEL A. FIGUEIREDO,
MARSHAL R. PATIL

Demand for bigger-screen TV receivers is growing. In the past, while efforts were made to increase the screen size of CRT TVs, because of limitations of the CRT glass tube technology it had not been possible to go beyond 102cm (40-inch) screens. As such, research work had been going on to develop screens bigger than possible with CRTs.

Liquid-crystal display (LCD) was the first to emerge as an alternative and initially found applications in display boards of various equipment, computer monitors, laptops, personal digital assistants, advertisement boards, etc. Plasma panels also allowed large-screen TV viewing. Both LCD and plasma TVs offer the advantage of a wider viewing angle. However, LCD TVs score over plasma TVs in that these consume lesser power and last longer.

Liquid crystal was discovered by Austrian botanist Fredrich Rheinizer in 1888. In mid 1960s, scientists discovered that applying an external electric charge could change the properties of liquid crystals, i.e., the luminance of light passing through them, and demonstrated the first liquid-crystal display.

A thin-film transistor liquid-crystal display (TFT-LCD) is a variant of LCD that offers improved image quality.

Structure of a TFT-LCD

TFT-LCDs have a sandwich-like structure with liquid crystal filled between two transparent glass electrode plates coated with indium tin oxide (ITO) as shown in Fig. 1.

Fig. 2 shows a colour TFT-LCD

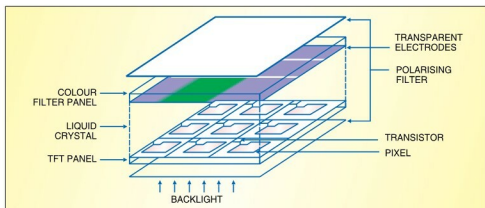


Fig. 1: TFT-LCD structure

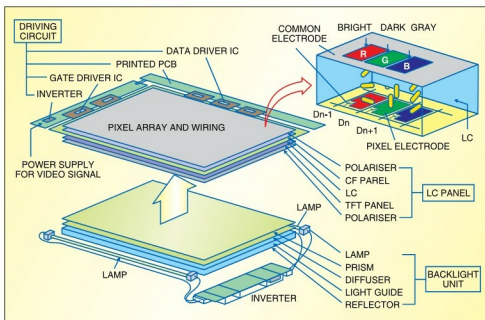


Fig. 2: Structure of TFT-LCD module

module. It consists of a TFT-LCD panel, a backlight unit and a driving unit (source driver and gate driver). The TFT-LCD panel has TFTs, liquid crystal, pixel electrodes, storage capacitors and colour filter. The colour filter has a colour resin film containing three primary colour pigments (red, green and blue).

As shown in Fig. 3, the backlight unit consists of a CCFL lamp with inverter, a light guide panel, a diffuser,

and bottom and top prisms. The light guide panel guides the light from the lamp. The diffuser uniformly diffuses the light received from the light guide. The prism concentrates the light and increases the luminance of light up to 1.55 times of the light guide.

Working principle

Liquid-crystal molecules are arranged randomly when there is no electric field